Force Sensor Data Greybox Modelling

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1 Experimental Data

1.1 Tooling and Coating Systems

A total of four PVD coating systems were investigated. These differ in both layer architecture and coating thickness:

- Monolayer coatings with thickness levels of 2.0 µm (Mono-20) and 3.5 µm (Mono-35),
- Bilayer coatings with thickness levels of 2.0 µm (Bilay-20) and 3.5 µm (Bilay-35).

The experiments were conducted across two independent test series (Reihe 1 and Reihe 2), resulting in a total of 41 individual tool samples.

1.2 Force Measurements

During each turning pass (referred to as *Drehen*), cutting forces were recorded at high temporal resolution using sensors. The measured force components are:

- Feed force: F_x,
- Thrust force: F_y,
- Cutting force: F_z.

These signals provide a continuous and high-frequency representation of the machining process and serve as the primary input features for the predictive model.

1.3 Wear Measurements

Tool wear is quantified by the flank wear land width VB, measured in micrometres (µm). Unlike the force signals, VB is not measured continuously. Instead, it is obtained through manual measurements at discrete cutting lengths.

Depending on the duration of each experimental run, between 5 and 20 wear measurements are available per tool sample.

This results in sometimes a bit more sparse and non-uniform distribution of target values along the tool life.

1.4 Data Challenges

- Force signals are available continuously for every turning pass,
- Wear measurements (VB) are sparse and irregularly spaced.

This discrepancy makes direct supervised learning challenging, as the model must infer the underlying wear progression from limited ground-truth observations.

2 Signal Preprocessing

The raw measurement data is stored in `.tdms` files. Each file contains the three force components F_x, F_y, and F_z, sampled at a fixed acquisition rate of approximately 6400 Hz. A single file corresponds to one complete turning pass ("Drehen"), capturing the full interaction between tool and workpiece, including a noise transition phases at tool entry and exit.

2.1 Downsampling

As an initial preprocessing step, the raw signals are downsampled by a factor of four. Specifically, every four consecutive samples are replaced by their arithmetic mean, reducing the effective sampling rate.

This step significantly reduces memory usage and computational cost while preserving the relevant signal characteristics, as tool wear evolves over relatively long time scales (seconds rather than milliseconds).

2.2 Threshold-Based Segmentation

The downsampled signals still contain regions where no cutting occurs, such as tool repositioning or idle motion. These regions are characterized by near-zero force values and must be removed to avoid contaminating the feature extraction process.

To isolate active cutting periods, a threshold-based segmentation is applied. A sample is classified as part of a cutting segment if at least one force component satisfies:

$$\max(|F_x|, |F_y|, |F_z|) > 150 \text{ N}.$$

Consecutive samples fulfilling this condition are grouped into segments. To ensure robustness against short signal fluctuations, small gaps between segments are bridged if their duration is less than 100 samples.

2.3 Edge Trimming

Even within detected cutting segments, the signal near the beginning and end is affected by transition noise, such as tool entry and exit effects. These regions often contain force spikes and oscillations that are not representative of steady-state cutting conditions.

To eliminate these artifacts, 1600 samples are removed from both the start and the end of each segment. At the downsampled rate of 1600 Hz, this corresponds to approximately one second of data at each boundary.

2.4 Signal Reconstruction

After segmentation and trimming, all remaining valid segments are concatenated in temporal order to form a single reconstructed signal for each turning pass.

The resulting signals

$$\tilde{F}_x(t), \quad \tilde{F}_y(t), \quad \tilde{F}_z(t)$$

represent clean, steady-state cutting conditions and have variable length depending on the duration of active cutting.

3 Feature Extraction

Following the signal preprocessing stage, each turning pass (Drehen) is represented by a clean force signal for all three axes. From these signals, a fixed set of statistical features is extracted independently for each force component F_x, F_y, and F_z.

This process gives us a total of 27 scalar features per Drehen (9 features per axis), resulting in a tabular representation suitable for machine learning.

3.1 Feature Computation per Axis

For each force signal $\tilde{F}(t)$, the following descriptors are computed:

- **Mean:**
$$\mu = \frac{1}{N}\sum_{i=1}^{N}\tilde{F}_i$$
Represents the average force level over the cutting pass.
- **Maximum:**
$$F_{\max} = \max_i \tilde{F}_i$$
Captures the peak force magnitude.
- **Minimum:**
$$F_{\min} = \min_i \tilde{F}_i$$
Describes the lower bound of force variation.
- **Standard Deviation:**
$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\tilde{F}_i - \mu)^2}$$
Measures the variability of the signal.
- **Skewness:**
$$\text{Skew} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{\tilde{F}_i - \mu}{\sigma}\right)^3$$
Quantifies the asymmetry of the force distribution.
- **Kurtosis (Fisher):**
$$\text{Kurt} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{\tilde{F}_i - \mu}{\sigma}\right)^4 - 3$$
Measures the tail heaviness relative to a Gaussian distribution.
- **Area Under Curve:**
$$A = \int \tilde{F}(t)\,dt$$
Approximated numerically using the trapezoidal rule, this represents the total force impulse over the cutting pass.
- **FFT Energy:**
$$E_{\text{FFT}} = \sum_k |\hat{F}_k|^2$$
where $\hat{F}_k$ denotes the discrete Fourier transform of the signal. This feature captures the total spectral energy.
- **Segment Length:**
$$N$$
The number of samples in the processed signal, serving as a proxy for the effective cutting duration.

3.2 Feature Matrix Structure

Each Drehen is represented as a single row in the feature matrix containing:

- The 27 extracted force features (9 per axis),

- The corresponding flank wear measurement VB (in µm),
- The cumulative cutting length s (in m),
- Properties of the coating, such as thickness, and others...,
- Metadata fields: `Sample_ID`, `Reihe`, `Schichtsystem`, `Run_ID`, and `Drehen_Index`.

4 Baseline Model: XGBoost (No Physics, No Interpolation)

As an initial baseline, a gradient-boosted decision tree model (XGBoost) is trained to predict the absolute flank wear VB (in µm) directly from the extracted force features described in Section 3.

- No physical constraints are imposed,
- No monotonicity is enforced on predictions,
- No interpolation or augmentation of wear measurements is performed.

4.1 Experimental Setup

The dataset is split at the sample level: 80% of runs are used for training and 20% for testing. This ensures that all measurements from a given run appear exclusively in either the training or test set.

Hyperparameters are optimised using grid search combined with 5-fold *GroupKFold* cross-validation, where folds are defined based on run number. Model selection is performed by minimising the mean absolute error (MAE) on VB.

Two feature configurations are evaluated:

Variant A - Without cutting length The cumulative cutting length s (Schnittweg) is excluded from the feature set. The model must infer the wear state solely from statistical properties of the force signals.

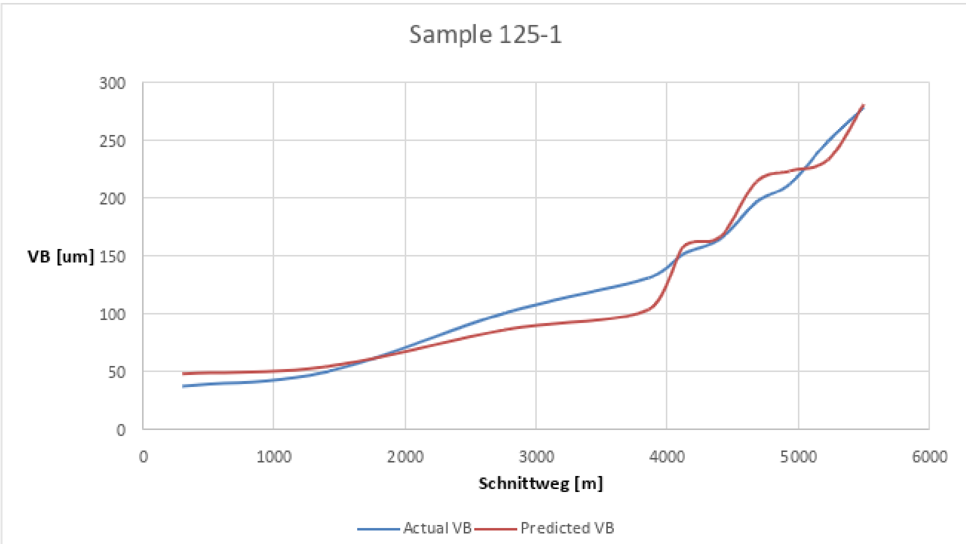

Figure 1: Prediction of flank wear VB using the XGBoost baseline model without cutting length (s) on sample 125-1. Results folder: "E:\Vorhersagen-ML\Werkstoff 1\Model Trainings\results-tree-models\XGBoost_PredictVB_NO-CLIP_split-RunNumber_NO-INTERPOLATION_NO-SCHNITTWEG_R2-87"

Variant B - With cutting length The cutting length s is included as an additional input feature. This provides the model with explicit information about the tool's progression along its life cycle.

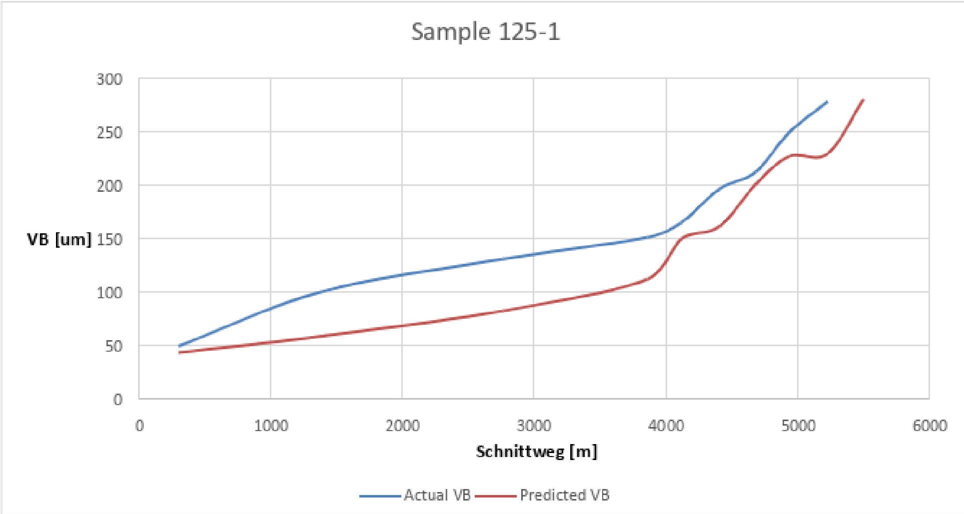

Figure 2: Prediction of flank wear VB using the XGBoost baseline model wiht cutting length (s) included as a feature. On sample 125-1. "E:\Vorhersagen-ML\Werkstoff 1\Model Trainings\results-tree-models\XGBoost_PredictVB_NO-CLIP_split-RunNumber_NO-INTERPOLATION_With-SCHNITTWEG_R2-92 "

4.2 Results

Variant	CV MAE (µm)	CV RMSE (µm)	CV R^2	Test MAE (µm)	Test RMSE (µm)	Test R^2
Without s	24.64	34.68	0.676	15.78	21.79	0.879
With s	18.41	26.48	0.800	12.89	17.49	0.922

Table 1: Performance of the XGBoost baseline model on cross-validation and held-out test data.

4.3 Discussion

The variant without cutting length achieves a test performance of $R^2 = 0.879$, demonstrating that the force signal alone contains significant information about tool wear.

However, a noticeable discrepancy exists between cross-validation performance ($R^2 = 0.676$) and test performance ($R^2 = 0.879$). This suggests that model performance varies across different runs and coating systems, and that the selected test split may contain comparatively easier-to-predict samples. (Maybe overfitted results.)

Including the cutting length as an input feature improves all evaluation metrics, with test performance increasing to $R^2 = 0.922$. This is expected, as s is strongly correlated with wear progression and provides direct information about the position along the tool life.

At the moment, the model can produce non-physical behaviour, such as decreasing wear between consecutive turning passes of the same tool. For this reason we seek to develop a physics-constrained models which is introduced in the following section.

5 Data Interpolation

5.1 Motivation

The goal of this was to increase the set of data we have for more effectively train ML models to predict tool wear. An important note is to make sure the number of newly interpolated datapoints is not to excessive as to not introduce redundancies.

The approach was such as to make sure the newly interpolated datapoints follow certain physical conditions, such as monotonicity. As wear would physically not decrease with respect to time. We used this idea when generating the new data, as this more certainly guarantees that the newly generated data would be physically meaningful.

5.1.1 Idea

The idea is to interpolate points by fitting a function to the current data we have per Sample_ID. But the constraint is that the function must always be positive.

5.1.2 Derivation of the Desired Function

Define a function $f(x)$ which is continuous and differentiable for all values of x. We will define it by first defining its first derivative.

$$f'(x) = (p_2x^2 + p_1x + p_0)^2$$

This derivative is positive for all values of p_2, p_1, p_0. Now we take the indefinite integral to find the actual function.

$$f'(x) = p_2^2x^4 + p_1^2x^2 + p_0^2 + 2p_1p_2x^3 + 2p_0p_2x^2 + 2p_0p_1x$$

Collecting the x^2 terms:

$$f'(x) = p_2^2x^4 + 2p_1p_2x^3 + (p_1^2 + 2p_0p_2)\,x^2 + 2p_0p_1\,x + p_0^2$$

Step 2: Integrate term by term

$$f(x) = \int f'(x)\,dx = \int \left[p_2^2x^4 + 2p_1p_2x^3 + (p_1^2 + 2p_0p_2)\,x^2 + 2p_0p_1\,x + p_0^2\right] dx$$

$$\int p_2^2\,x^4\,dx = \frac{p_2^2}{5}\,x^5$$

$$\int 2p_1p_2\,x^3\,dx = \frac{p_1p_2}{2}\,x^4$$

$$\int (p_1^2 + 2p_0p_2)\,x^2\,dx = \frac{p_1^2 + 2p_0p_2}{3}\,x^3$$

$$\int 2p_0p_1\,x\,dx = p_0p_1\,x^2$$

$$\int p_0^2\,dx = p_0^2\,x$$

Step 3: Final result

$$\boxed{f(x) = \frac{p_2^2}{5}\,x^5 + \frac{p_1 p_2}{2}\,x^4 + \frac{p_1^2 + 2p_0 p_2}{3}\,x^3 + p_0 p_1\,x^2 + p_0^2\,x + f_0}$$

We now have a total number of 5 prefactors, but 4 unique variable parameters: p_2, p_1, p_0, f_0. The parameter f_0 sets the initial wear at $x = 0$, i.e. $f(0) = f_0$.

Global parameter fitting

The four parameters $\boldsymbol{\theta} = (f_0, p_0, p_1, p_2)$ are fitted **globally** over the entire wear curve using nonlinear least squares:

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{N} \big(f(x_i;\, \boldsymbol{\theta}) - \mathrm{VB}_i\big)^2$$

A fixed anchor point $(x, \mathrm{VB}) = (0, 0)$ is prepended to each sample's data before fitting, ensuring that the curve starts at zero wear.

Generating new interpolated points

Once $\hat{\boldsymbol{\theta}}$ is found, $f(x)$ can be evaluated at any arbitrary cutting length. Given a desired number of points M per measurement interval $[x_i,\, x_{i+1}]$, a uniform grid is placed over each interval and f is evaluated at each grid point:

$$x_{i,k} = x_i + \frac{k}{M}(x_{i+1} - x_i), \qquad k = 0, 1, \ldots, M-1$$

$$\widehat{\mathrm{VB}}_{i,k} = f\Big(x_{i,k};\, \hat{\boldsymbol{\theta}}\Big)$$

The number of interpolated points M per interval is a free hyperparameter and can be chosen arbitrarily. Because f is a single smooth function fitted over the whole curve, the interpolated points are globally consistent. They all lie on the same fitted wear trajectory rather than being piecewise independent fits.

6 Using Interpolated Data and Physical Postprocessing

This section introduces two improvements: dataset enlargement via interpolation and the incorporation of physical constraints through incremental modelling.

6.1 Dataset Enlargement via Monotone Interpolation

To increase the effective dataset size, a monotone wear model is fitted to each sample's measured (s, VB) pairs. Based on this fitted curve, 40 additional interpolated points are generated within each measurement interval.

The XGBoost model is then retrained on this augmented dataset to predict VB directly, maintaining the same target formulation as in the baseline model.

Model	CV MAE (µm)	CV RMSE (µm)	CV R^2	T MAE (µm)	T RMSE (µm)	T R^2
Baseline XGBoost	18.41	26.48	0.800	12.89	17.49	0.922
XGBoost+interpolation	16.80	22.71	0.826	15.66	19.61	0.922

Table 2: Effect of dataset enlargement via interpolation on model performance.

The cross-validation performance improves slightly ($R^2 = 0.826$ vs. 0.800), indicating more consistent generalisation across folds. However, the test performance remains unchanged ($R^2 = 0.922$), as the interpolated points do not introduce new information but rather provide a smoother and denser representation of the existing data.

6.2 Predicting Incremental Wear (ΔVB) with Monotonicity Clipping

A physical property of tool wear is that it is monotonically non-decreasing. The direct prediction of VB does not enforce this constraint.

To address this, the prediction target is changed from absolute wear VB to incremental wear:

$$\Delta VB_i = VB_i - VB_{i-1}.$$

The model is trained to predict ΔVB for each turning pass. The full wear trajectory is then reconstructed via cumulative summation, based at the initial wear value:

$$\widehat{VB}_i = VB_0 + \sum_{k=1}^{i} \widehat{\Delta VB}_k.$$

To enforce physical meaning, a post-processing step is applied in which negative predicted values are clipped to zero:

$$\widehat{\Delta VB}_i \leftarrow \max\left(0, \widehat{\Delta VB}_i\right).$$

This guarantees that the reconstructed wear curve is monotonically increasing, regardless of individual prediction errors.

6.2.1 Results of Tree Models with Interpolation and Postprocessing

In this section different types of trained XGBoost models' results are shown. Parameters that differ per training are: Number of interpolated points (40 or 3 per turning pass); Direct prediction of VB / Prediction of ΔVB with VB curve reconstruction and postprocessing; Cuttinglength included or not included during the training process.

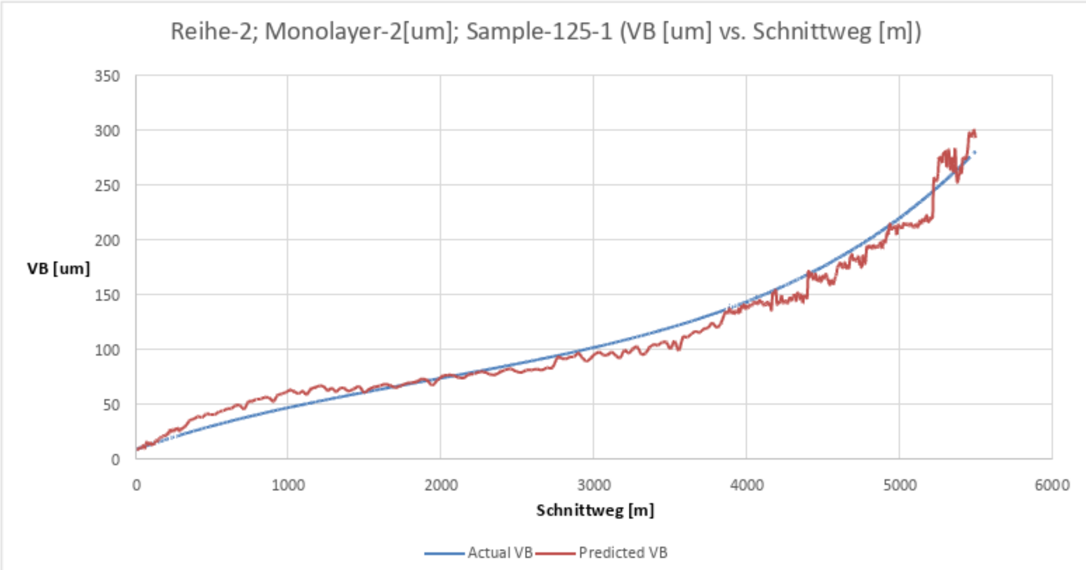

Figure 3: Prediction of flank wear VB using the XGBoost baseline model wiht cutting length (s) included as a feature. On sample 125-1. **40 newly interpolated intermediate points**. No post processing after predictions; **Direct VB value predictions**. **Schnittweg is included as a training feature**. Path: "E:\Vorhersagen-ML\Werkstoff 1\Model Trainings\results-tree-models\AllTreeModels_PredictVB_split-RunNumber_Has-Schnittweg_NOCLIP_R2-92"

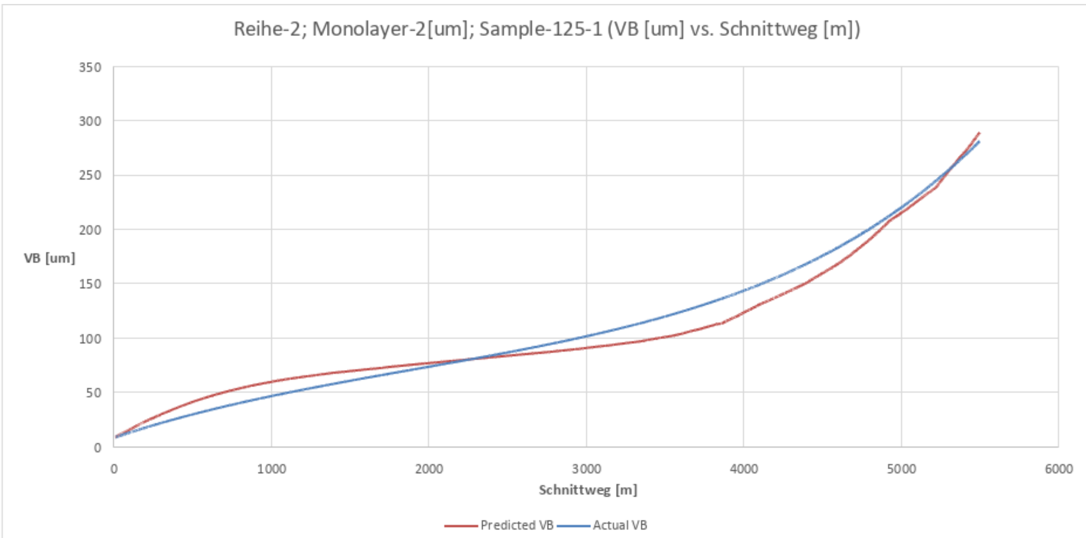

Figure 4: Prediction of flank wear VB using the XGBoost baseline model wiht cutting length (s) included as a feature. On sample 125-1. **40 newly interpolated intermediate points**. Includes a post processing clipping function after predictions; **Wear increment predictions**, used to reconstruct the VB signal. **Schnittweg is included as a training feature**. Path: "E:\Vorhersagen-ML\Werkstoff 1\Model Trainings\results-tree-models\XGBoost_PredictDeltaVB_clip_split-RunNumber_Has-Schnittweg_R2-94"

Model	T MAE (µm)	T RMSE (µm)	Test R^2
XGBoost+interpolation (direct VB)	15.66	19.61	0.922
XGBoost+interpolation (ΔVB + clipping)	15.30	17.28	0.940

Table 3: Performance improvement achieved by predicting incremental wear with monotonicity enforcement.

Reformulating the problem in terms of incremental wear leads to consistent improvements across all evaluation metrics. The test R^2 increases from 0.922 to 0.940, while the RMSE decreases from 19.6 μm to 17.3 μm.

This improvement does not arise from additional information in the data, but from a better alignment between the model formulation and the underlying physical process. By predicting wear increments and enforcing non-negativity, the model is constrained to produce physically plausible wear trajectories.

Below there results for the models are shown:

- XGBoost ; 3 interpolated ; with cuttinglength ; directVB
- XGBoost ; 3 interpolated ; without cuttinglength ; directVB
- XGBoost ; 3 interpolated ; with cuttinglength ; deltaVB
- XGBoost ; 3 interpolated ; without cuttinglength ; deltaVB

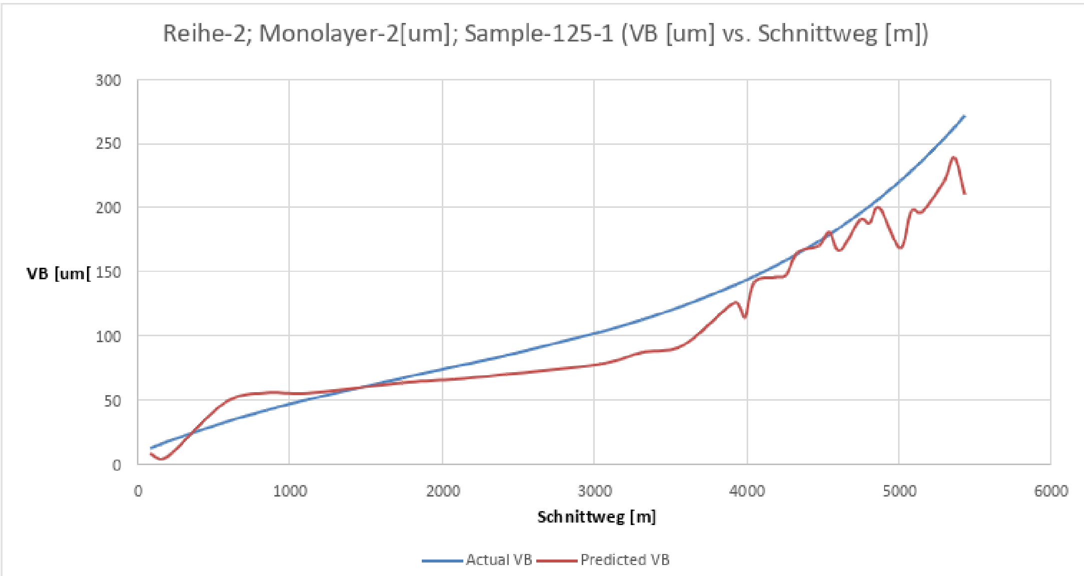

Figure 5: Prediction of flank wear VB using the XGBoost model. On sample 125-1. **3 newly interpolated intermediate points**. **Direct VB value predictions**. **Schnittweg is NOT included as a training feature**. Path: "E:\Vorhersagen-ML\Werkstoff 1\Model Trainings\results-tree-models\XGBoost_PredictVB_NO-CLIP_split-RunNumber_WITH-3-POINT-INTERPOLATION_NO-SCHNITTWEG_R2-87"

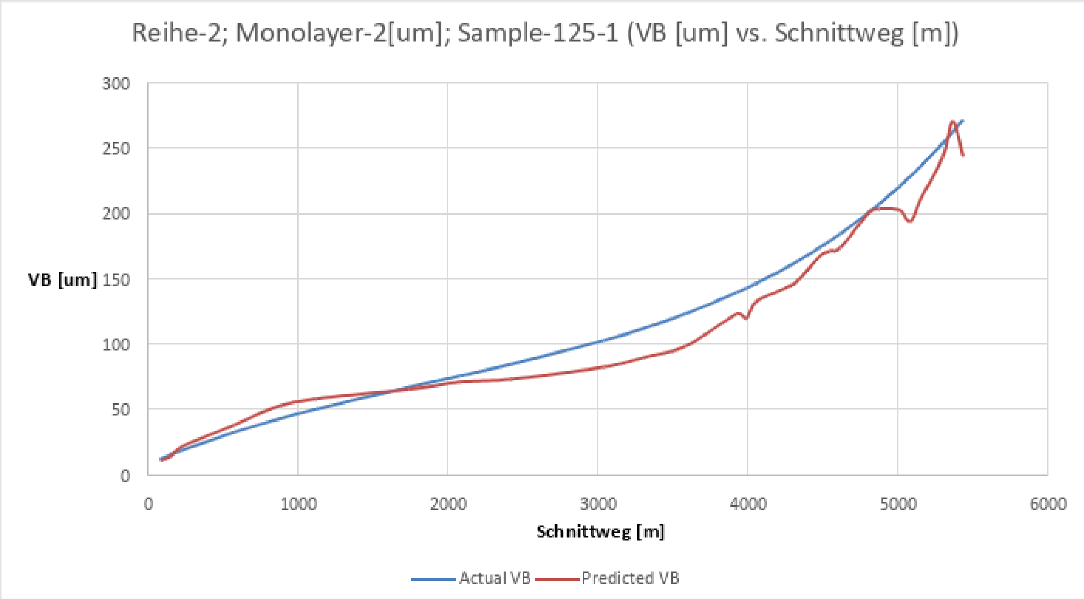

Figure 6: Prediction of flank wear VB using the XGBoost model wiht cutting length (s) included as a feature. On sample 125-1. **3 newly interpolated intermediate points**. **Direct VB value predictions**. **Schnittweg IS included as a training feature**. Path: "E:\Vorhersagen-ML\Werkstoff 1\Model Trainings\results-tree-models\XGBoost_PredictVB_NO-CLIP_split-RunNumber_WITH-3-POINT-INTERPOLATION_WITH-SCHNITTWEG_R2-93"

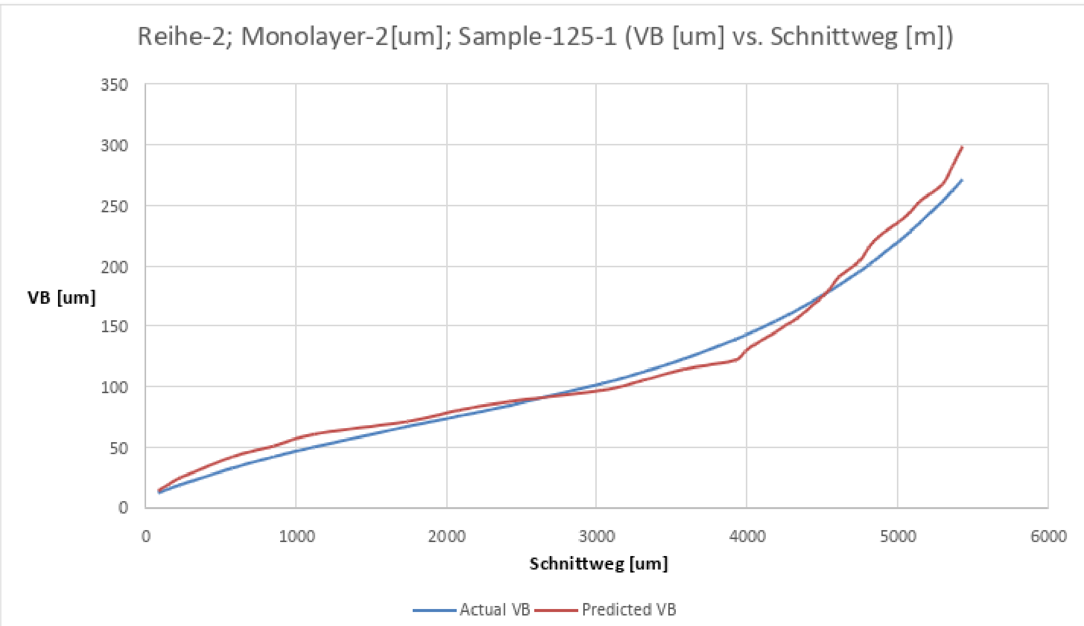

Figure 7: Prediction of flank wear VB using the XGBoost model. On sample 125-1. **3 newly interpolated intermediate points**. Includes a post processing clipping function after predictions; **Wear increment predictions**, used to reconstruct the VB signal. **Schnittweg is NOT included as a training feature**. Path: "E:\Vorhersagen-ML\Werkstoff 1\Model Trainings\results-tree-models\XGBoost_PredictDeltaVB_WITH-CLIP_split-RunNumber_WITH-3-POINT-INTERPOLATION_NO-SCHNITTWEG_R2-93"

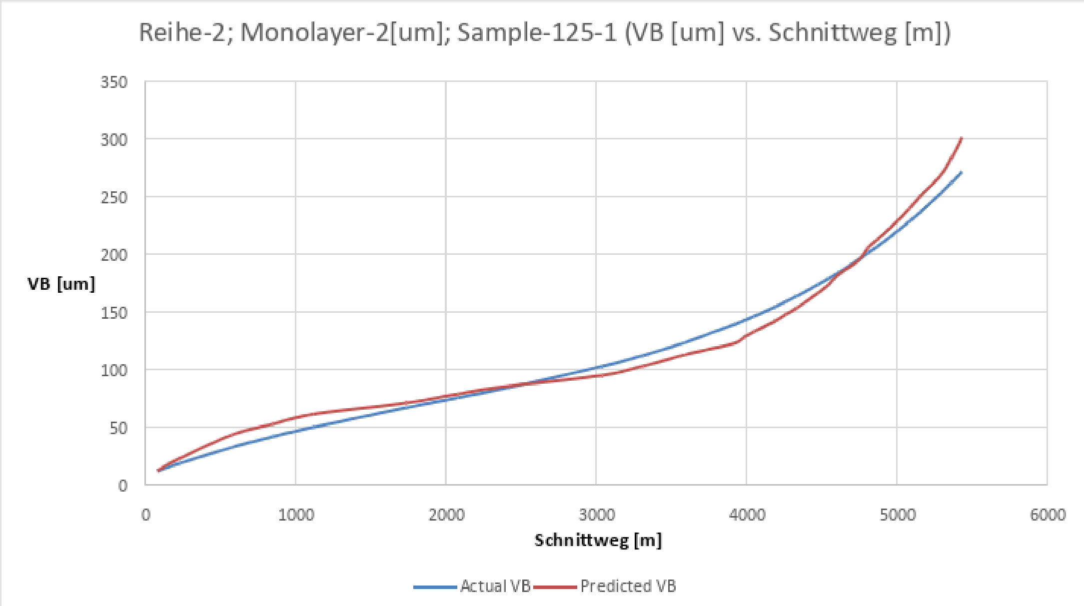

Figure 8: Prediction of flank wear VB using the XGBoost model with cutting length (s) included as a feature. On sample 125-1. **3 newly interpolated intermediate points**. Includes a post processing clipping function after predictions; **Wear increment predictions**, used to reconstruct the VB signal. **Schnittweg IS included as a training feature**. Path: "E:\Vorhersagen-ML\Werkstoff 1\Model Trainings\results-tree-models\XGBoost_PredictDeltaVB_WITH-CLIP_split-RunNumber_WITH-3-POINT-INTERPOLATION_WITH-SCHNITTWEG_R2-93.4"

Model	CV MAE (µm)	CV RMSE (µm)	CV R^2	T MAE (µm)	T RMSE (µm)	T R^2
XGBoost, no s	23.02	31.31	0.692	18.27	22.59	0.867
XGBoost, with s	14.74	19.92	0.882	12.38	15.47	0.938

Table 4: XGBoost predicting VB directly with 3-point interpolation, split by sample ID. s denotes cutting length (Schnittweg).

Model	CV MAE (µm)	CV RMSE (µm)	CV R^2	T MAE (µm)	T RMSE (µm)	T R^2
XGBoost ΔVB, no s	2.39	3.48	0.436	13.26	16.56	0.929
XGBoost ΔVB, with s	1.79	2.70	0.660	12.16	15.96	0.934

Table 5: XGBoost predicting ΔVB with post-processing clip and VB reconstruction via cumulative sum, with 3-point interpolation, split by sample ID. **CV metrics are computed on** ΔVB; test metrics are on reconstructed VB. s denotes cutting length (Schnittweg).

7 Motivation for a Physics-Constrained Neural Network

7.1 Reducing Dependence on Interpolation

The interpolation strategy used in Section 5 significantly increases the number of training samples by generating synthetic points along each wear trajectory. In the extreme case (e.g., 40 interpolated points per interval), the majority of the dataset consists of interpolated values rather than actual measurements.

Although this improves coverage of the wear curve, it introduces a potential bias: the model is effectively trained on smooth evaluations of a fitted function. As a result, it may learn the shape of the interpolation rather than the underlying relationship between force signals and wear.

To rduce this risk, the neural network model uses a reduced interpolation density of only three additional points per measurement interval.

7.2 Limited Generalisation

The evaluation of the XGBoost models is performed on a held-out test set split by run number. However, all samples originate from the same four coating systems and the same workpiece material as the training data.

In practical applications, a wear prediction model must generalise to previously unseen conditions, such as new coating systems, materials, or cutting parameters.

It could very well be the case that the previously shown models wont perform well against unseen systems, due to our limited training data and currently limited physics-based constraints. And the current physics based constraints which are used and described above are not part of the model flow. The model does not train itself based on those constraints, but rather we introduce them after the model has done its job.

7.3 Insufficient Incorporation of Physical Knowledge

In the tree-based pipeline, physical consistency is enforced only through post-processing, namely by clipping negative wear increments:

$$\widehat{\Delta VB}_i \leftarrow \max(0, \widehat{\Delta VB}_i).$$

Yes, this ensures monotonicity of the reconstructed wear curve, however it does not encode any knowledge about the expected shape of the wear progression.

Tool wear in turning typically follows a three-phase pattern:

- Rapid initial wear (break-in phase),
- Approximately linear steady-state wear,
- Accelerated wear in the catastrophic phase.

This structure reflects the underlying physical processes and is largely independent of specific experimental conditions. A model that does not use this knowledge for its training may produce physically implausible wear trajectories for unseen systems, even if its numerical error is low.

7.4 Motivation for an End-to-End Differentiable Model

Another key limitation of the XGBoost-based approach is that physical constraints are applied as a post-processing step. This means that the learning algorithm is not aware of these constraints during training.

A neural network allows the integration of physical constraints directly into the model architecture and loss function. By designing the model such that monotonicity is enforced by construction, and by incorporating physics-informed loss terms, the entire pipeline becomes end-to-end differentiable.

This enables gradients to propagate through all components of the model, ensuring that the optimisation process is guided by both data and physical principles.

7.5 Summary

The transition to a neural network-based approach is therefore motivated by three key factors:

- Reducing reliance on synthetic interpolated data,
- Improving generalisation beyond the training distribution,
- Embedding physical knowledge directly into the model and training process.

8 Phase-Aware NN Model Formulation

Instead of predicting the wear value $VB(t)$ directly, the model predicts the incremental wear (or wear rate) at each step. Given input features $\mathbf{x}_t$, a neural network f_θ outputs an unconstrained value:

$$z_t = f_\theta(\mathbf{x}_t)$$

To ensure physical consistency, this output is mapped to a non-negative wear rate using the softplus function:

$$r_t = \text{softplus}(z_t)$$

The actual wear increment is then computed as:

$$\Delta VB_t = r_t \cdot \Delta s_t$$

where Δs_t is the cutting length increment. The total wear is reconstructed via cumulative summation:

$$VB(t) = VB_0 + \sum_{i=1}^{t} \Delta VB_i$$

This formulation guarantees:

$$\Delta VB_t \geq 0 \quad \Rightarrow \quad VB(t) \text{ is monotonically increasing}$$

8.1 Positivity constraint via shifted softplus

To ensure that the predicted wear increments are non-negative, the raw network output z_t is mapped through a smooth activation function. A natural choice is the softplus function:

$$\text{softplus}(z) = \log\left(1 + e^z\right),$$

which satisfies $\text{softplus}(z) > 0$ and is fully differentiable.

However, a standard softplus has a strictly positive output even for large negative inputs:

$$\text{softplus}(0) \approx 0.693,$$

meaning the model would predict a non-zero wear rate even when the underlying signal suggests no wear.

To address this, a **shifted softplus** is used:

$$r_t = \max\left(0, \text{softplus}(z_t) - \epsilon\right),$$

where $\epsilon > 0$ is a small constant.

This modification ensures that the predicted wear rate can approach zero for sufficiently negative inputs, while still maintaining smoothness in the region where learning is active.

Importantly, the softplus function is continuously differentiable:

$$\frac{d}{dz}\text{softplus}(z) = \sigma(z) = \frac{1}{1+e^{-z}},$$

which guarantees stable gradient flow during backpropagation.

The use of a smooth, differentiable positivity constraint is essential: unlike hard thresholding functions (e.g., $\max(0, z)$), it allows gradients to propagate through the entire model, enabling end-to-end optimisation of both the wear rate and the reconstructed wear trajectory.

8.2 Baseline Loss Function

The initial model is trained using a standard mean squared error (MSE) loss on the reconstructed wear:

$$\mathcal{L}_{VB} = \text{MSE}(VB_{\text{pred}}, VB_{\text{true}})$$

While this ensures accurate prediction of the cumulative wear, it does not determine the wear rate trajectory. Multiple different rate curves can integrate to the same VB, leading to physically non-accurate solutions.

8.3 Increment Supervision

To address this ambiguity, an additional loss term is introduced on the wear increments:

$$\mathcal{L} = \mathcal{L}_{VB} + \delta \cdot \text{MSE}(\Delta VB_{\text{pred}}, \Delta VB_{\text{true}})$$

This directly punishes the rate prediction, improving both convergence and physical accuracy.

8.4 Regularization of the Wear Rate (NOT USED IN THE FINAL RESULT BELOW)

To help enforce a constraint regarding the typical s-curve shape of a wear vs cutting length graph, an 2 extre term were added:

Smoothness constraint

$$\mathcal{L}_{\text{smooth}} = \sum_t (\Delta VB_{t+1} - \Delta VB_t)^2$$

This is supposed to discourage unrealistic spikes in wear rate.

Convexity (U-shape) constraint

$$\mathcal{L}_{\text{u-shape}} = \sum_t \max(0, -\nabla^2 r_t)^2$$

This encourages the wear rate to follow a convex trend, reflecting the general behaviour of tool wear.

8.5 Phase-Aware Shape Constraints

Tool wear typically follows three distinct phases:

- **Phase 1 (logarithmic-like):** rapid initial wear, well approximated by a logarithmic curve
- **Phase 2 (linear):** approximately linear wear progression
- **Phase 3 (catastrophic / exponential):** accelerating wear, approximated by an exponential curve

To explicitly encode this, I introduced phase-specific losses. For each phase, the predicted VB values are fitted to the expected functional form using differentiable least squares:

$$\begin{aligned} &\text{Phase 1:} \quad VB \approx a \log(s+1) + b \\ &\text{Phase 2:} \quad VB \approx as + b \\ &\text{Phase 3:} \quad \log(VB) \approx as + b \end{aligned}$$

The deviation from these fitted curves is penalized:

$$\mathcal{L}_{\text{phase}} = \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_2 + \lambda_3 \mathcal{L}_3$$

What is important, the least-squares fitting is performed inside the computational graph, so gradients propagate through the fit parameters back to the model.

This means the model must produce predictions that match the expected physical behaviour.

8.6 Final Objective

The complete training objective becomes:

$$\mathcal{L} = \mathcal{L}_{VB} + \delta \mathcal{L}_{\Delta VB} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}} + \lambda_{\text{u}} \mathcal{L}_{\text{u-shape}} + \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_2 + \lambda_3 \mathcal{L}_3$$

Note: For the final result which is shown later, λ_{smooth} and λ_{u} have been set to 0.

8.7 Key Insight

The central idea is that:

> *Physical constraints should be enforced by the model structure whenever possible, and refined by the loss function where necessary.*

By predicting wear increments instead of absolute wear, enforcing positivity, and incorporating phase-specific knowledge, the model becomes more physically interpretable.

9 Final Physics-Aware NN Model Results

9.1 Base NN. Direct VB Prediction. No Phase Awareness

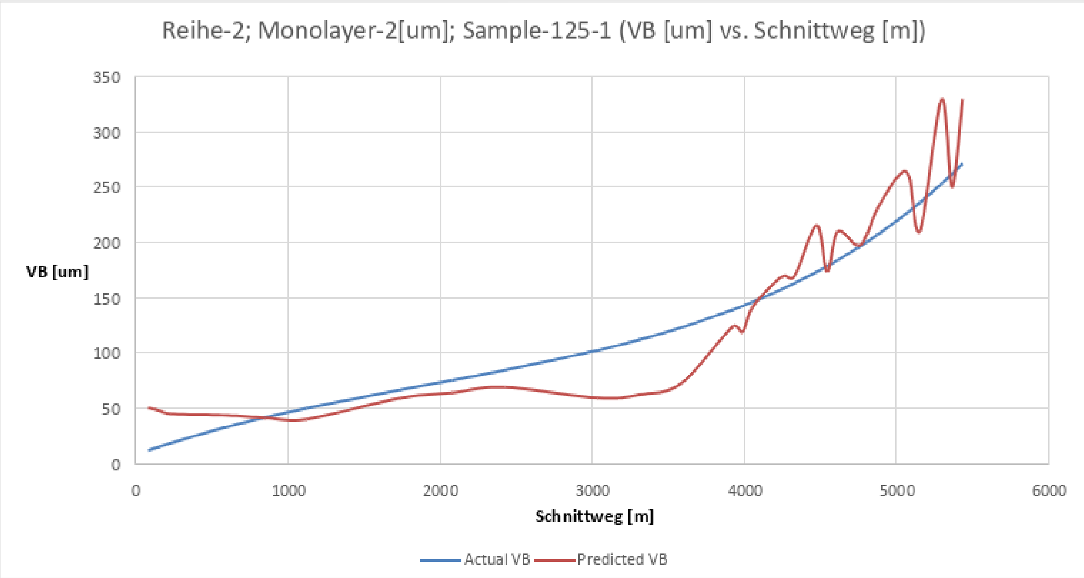

Figure 9: Prediction of flank wear VB using a base MLP NN model without cutting length (s) included as a feature. On sample 125-1. 3 newly interpolated intermediate points. Path: "E:\Vorhersagen-ML\Werkstoff 1\Model Trainings\results-nn-predVB-3-interpolated-points\run_20260416_102156 "

Model	**CV MAE** (µm)	**CV RMSE** (µm)	**CV R^2**	**T MAE** (µm)	**T RMSE** (µm)	**T R^2**
MLP (direct VB)	–	–	–	20.88	27.55	0.802

Table 6: Performance of a feedforward neural network predicting VB directly without physical constraints.

Hyperparameter	**Value**
Data split	Sample ID
Epochs (max)	120
Best epoch	87
Batch size	64
Hidden layers	2
Hidden units	128
Dropout	0.1
Optimizer	Adam
Learning rate	5×10^{-4}
Weight decay	1×10^{-4}
Loss function	MSE on VB
Interpolation density	3 points per interval

Table 7: Training configuration of the MLP model predicting VB directly.

9.2 NN Model. Ware Rate Prediction. Softplus. No Phase Awareness

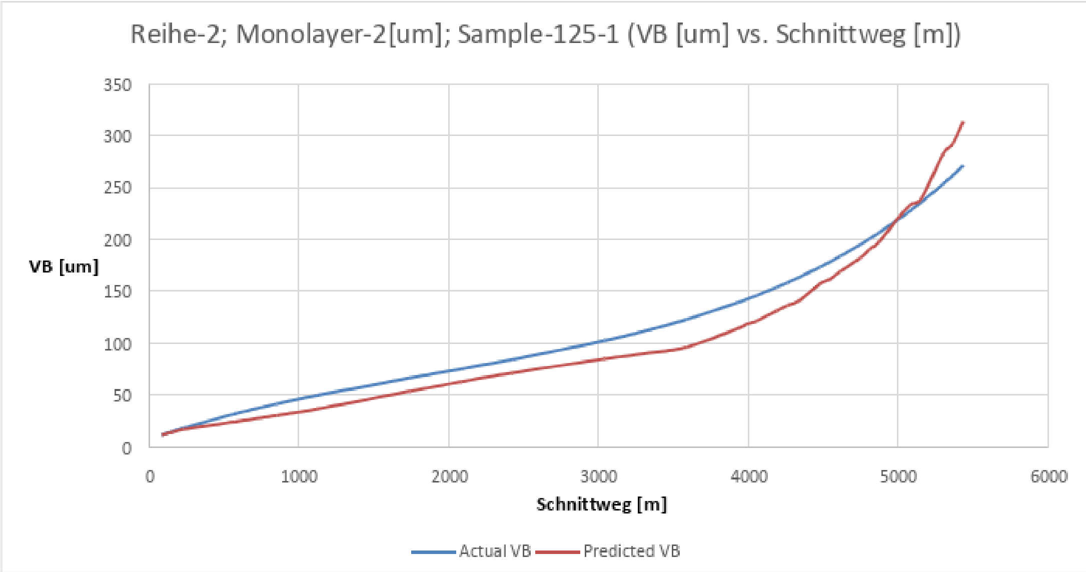

Figure 10: Prediction of flank wear VB using an MLP-based model. MLP outputs wear rate, wear rate get converted to incremental wear and then used to reconstruct the VB curve. **No Phase Awareness** Without cutting length (s) included as a feature. On sample 125-1. 3 newly interpolated intermediate points. Path: "E:\Vorhersagen-ML\Werkstoff 1\Model Trainings\results-nn-predVB-3-interpolated-points\run_20260416_102156 "

Model	**CV MAE** (µm)	**CV RMSE** (µm)	**CV** R^2	**T MAE** (µm)	**T RMSE** (µm)	**T** R^2
Monotone NN (ΔVB)	–	–	–	18.51	22.87	0.864

Table 8: Performance of the monotone neural network predicting incremental wear (ΔVB) with architectural enforcement of monotonicity.

Hyperparameter	Value
Data split	Sample ID
Epochs (max)	120
Best epoch	44
Batch size	16
Hidden units	128
Dropout	0.1
Optimizer	Adam
Learning rate	5×10^{-4}
Weight decay	1×10^{-4}
Loss function	MSE ($VB + \Delta VB$)
$\lambda_{\Delta VB}$	0.1
Phase losses	Disabled
Positivity constraint	Shifted softplus
Shift value (ϵ)	10^{-3}
Interpolation density	3 points per interval

Table 9: Training configuration of the monotone neural network predicting wear increments.

9.3 Phase Label Feature

To enable phase-specific supervision, an additional phase label is introduced into the feature matrix. Each turning pass (Drehen) is assigned to one of three phases based on its relative position along the wear trajectory of the corresponding tool sample:

- **Phase 1 (break-in):** early stage with rapid wear and decreasing wear rate,
- **Phase 2 (steady-state):** intermediate stage with approximately linear wear progression,
- **Phase 3 (catastrophic):** final stage with accelerating wear toward tool failure.

The phase labels are manually inserted for each sample during the feature matrix construction. During training, these labels are provided to the model to activate the corresponding phase-specific loss terms only on the relevant data points.

9.4 Training Configuration

The neural network is trained using the combined loss function described in Section 6. The weighting coefficients for the individual loss components are set as follows:

Parameter	Value
$\lambda_{\Delta VB}$	0.1
λ_{phase1}	0.1
λ_{phase2}	0.1
λ_{phase3}	0.1
λ_{smooth}	0.0
$\lambda_{\text{u-shape}}$	0.0

Table 10: Loss weighting configuration for the phase-aware neural network.

The smoothness and convexity (u-shape) regularisation terms are disabled in this configuration. Training is performed with early stopping (patience of 20 epochs), terminating at epoch 44 out of a maximum of 120 epochs, with a best validation loss of 262.4.

9.5 Results

Model	Test MAE (µm)	Test RMSE (µm)	Test R^2
Phase-Aware Neural Network	18.51	22.88	0.864

Table 11: Performance of the phase-aware neural network.

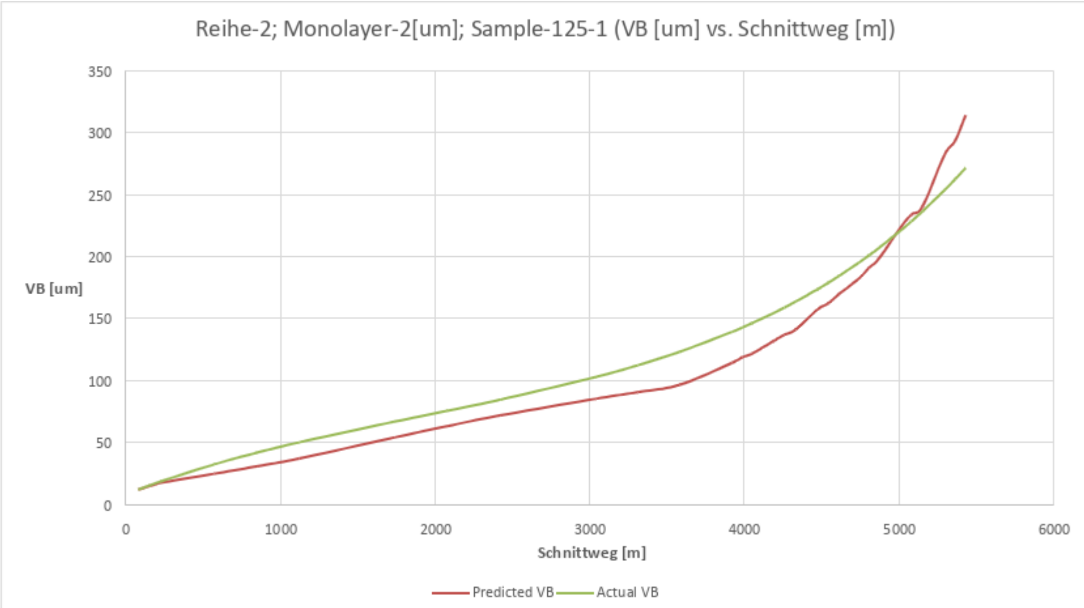

Figure 11: Prediction of flank wear VB using the Physics-Informed NN model without cutting length (s) included as a feature. On sample 125-1. 3 newly interpolated intermediate points. Path: "E:\Vorhersagen-ML\Werkstoff 1\Model Trainings\results-nn-phase-aware-3-interpolated-points\20260317_PhaseAware_split-SampleID_delta01_phase-all01_R2-86 "

The model achieves a test performance of $R^2 = 0.864$. The data split is performed at the sample (run) level, ensuring that complete wear trajectories are held out during training.

9.6 Discussion

Compared to the XGBoost models presented in Section 5, the neural network achieves a lower R^2 score. However, both approaches are evaluated under the same data split.

The key difference lies not in the numerical performance alone, but in the structure of the predictions. The XGBoost models optimise purely for predictive accuracy and require post-processing to enforce monotonicity. The neural network incorporates physical constraints directly into both the model architecture and the loss function.

As a result, the neural network produces wear trajectories that are **inherently physically consistent**. The predicted curves follow the expected three-phase behaviour (break-in, steady-state, catastrophic) and do not exhibit non-physical behaviours such as decreasing wear or discontinuities (**And this enforcement will be true even for unseen systems**).

This highlights an important trade-off: the XGBoost models achieve higher accuracy on the given dataset, but the neural network provides stronger guarantees of physical consistency and a more structured representation of the underlying wear process.

In applications where interpretability, robustness, and generalisation are critical, such physics-informed modelling approaches offer a significant advantage despite slightly lower predictive performance.

The next step would be to try and evaluate those models with data from new unseen coating systems.